

P-725.xCDE1S

Scanner System for Microscope Objectives

Dynamic focus scanner of the entry-level class for travel ranges up to 400 μm , incl. controller



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User Manual

P725UM0003DE, applies to P-725.xCDE1S
CBo, 07/23/2025



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About this Document

This user manual provides you with the information required for using the P-725.xCDE1 (also referred to as "P-725", "focus scanner", or "positioner"). The P-725.xCDE1 is part of P-725.xCDE1S systems (p. 8). The letter "x" in the product number represents the different models.

Basic knowledge of control technology, drive technologies, and suitable safety measures is assumed.

Typographic Conventions

Symbol/ Label	Meaning
1.	Action consisting of several steps with strict sequential order
2.	
➤	Action consisting of one or more steps without relevant sequential order
▪	Bullet point
p. 5	Cross-reference to page 5

Symbols Used

CAUTION



Dangerous situation

Failure to comply could result in minor injuries or damage the equipment.

- Precautions to avoid the risk.

NOTICE



Dangerous situation

Failure to comply could result in damage to the equipment.

- Precautions to avoid the risk.

INFORMATION

Additional information that can affect your application.

Symbol/Labeling



Meaning

Warning signs attached to the product which refer to detailed information in this manual.

"Residual Voltage" warning sign: Risk of electric shock (p. 6) for models with D-sub connector

Figures

For better clarity, the illustrations can vary in color, proportions, and level of detail compared to actual circumstances. Photographic illustrations may also differ and must not be seen as guaranteed properties.

Other Applicable Documents

Product	Document
P-5xx / P-6xx / P-7xx piezo positioners	PZ240EK Short Instructions

Downloading Manuals

INFORMATION

If a manual is missing or problems occur with downloading:

- Contact our customer service department (p. 23).

Downloading manuals

1. Open the website **www.pi.ws**.
2. Search the website for the product number (e.g., P-725).
3. In the search results, select the product to open the product detail page.
4. Select **Downloads**.
Manuals are shown under **Documentation**. Software manuals are shown under **General Software Documentation**.
5. For the desired manual, select **ADD TO LIST** and then **REQUEST**.
6. Fill out the request form and select **SEND REQUEST**.
The download link will be sent to the email address entered in the form.

Safety

Intended Use

The P-725 focus scanner is a laboratory device according to DIN EN 61010-1. It is intended for indoor use and use in an environment free from dirt, oil, and lubricants.

According to its design, the P-725 is intended for positioning and shifting microscope objectives on one axis.

The P-725 can be mounted horizontally or vertically. The specifications refer to a P-725 with vertically aligned motion axis.

The P-725 can only be used as intended when fully mounted and connected. It must be operated with the controller included in the scope of delivery (p. 10).

The P-725 may only be started, operated, maintained, and cleaned by authorized and qualified staff.

Safety Precautions

CAUTION



Dangerous voltage and residual charge in piezo actuators!

The P-725 is driven by piezo actuators. Temperature changes and compressive stress can induce charges in piezo actuators. Piezo actuators can remain charged for several hours after disconnecting the electronics. Touching or short-circuiting the contacts in the P-725's connector could lead to minor injuries from electric shock. Abrupt contractions may also destroy the piezo actuators.

- Do **not** open the P-725.
- Discharge the piezo actuators before installing (p. 20).
- Do **not** pull out the connector of the electronics during operation.



Touching the contacts in the plug connector can lead to an electric shock (max. 120 V DC) and minor injuries.

- Do **not** touch the contacts in the connector.
- Secure the P-725's connector with screws to prevent it from being pulled out of the controller.

CAUTION



Risk of electric shock if the protective earth conductor is not connected!

If the protective earth conductor is not connected or is not connected properly, hazardous touch voltages can occur on the P-725 during system malfunction or failure. Touching the P-725 under these circumstances may result in minor injury due to electric shock.

- Connect the P-725 to a protective earth conductor (p. 12) before use.
- Do **not** disconnect the protective earth conductor during operation.
- If the protective earth conductor needs to be disconnected temporarily (e.g., during modifications), make sure that the P-725 is reconnected to the protective earth conductor before restarting.

NOTICE



Parts falling down when removing the thread adapters!

While removing the upper thread adapter (microscope side), the P-725 can drop if it is not secured properly or screwed tightly. While removing the lower thread adapter (objective side), the thread adapter and the screwed in objective might fall.

- Secure all parts from falling when the thread adapters are removed:
 - Screw the objective out before disassembly.
 - When you loosen the upper thread adapter, hold the P-725 tightly.

NOTICE



Unsuitable cables!

Unsuitable cables can damage the electronics.

- Only use cables from PI for connecting the P-725 to the electronics.

INFORMATION

Cable extensions can affect the performance of the P-725.

- Do not use extension cables. If you need longer cables, contact our customer service department (p. 23).

Product Description

Model Overview

Model	Description
P-725.1CDE1S	PIFOC scanner system for microscope objectives; fast step-and-settle; 100 µm travel range; capacitive, direct position measuring; D-sub 7W2 (m) connector; 1.5 m cable length. System consisting of P-725.1CDE1 PIFOc scanner and E-709.1C1L controller.
P-725.4CDE1S	PIFOC scanner system for microscope objectives; fast step-and-settle; 400 µm travel range; capacitive, direct position measuring; D-sub 7W2 (m) connector; 1.5 m cable length. System consisting of P-725.4CDE1 PIFOc scanner and E-709.1C1L controller.

Thread Adapters and Spacers

The P-725.xxM and P-725.xxL thread adapters are available with apertures that have different inner diameters:

Thread size	Microscope side thread adapter		Objective / lens side thread adapter	Objective spacer for correcting the parfocal length
	P-725.xxM	ID in mm	P-725.xxL	P-725.xxS
M26 x 0.75	P-725.02M	22	P-725.02L	P-725.02S
M27 x 0.75	P-725.03M	22	P-725.03L	P-725.03S
M28 x 0.75	P-725.04M	22	P-725.04L	P-725.04S
M32 x 0.75	P-725.05M	25	P-725.05L	P-725.05S
M26 x 1/36"	P-725.06M	22	P-725.06L	P-725.06S
M19 x 0.75	P-725.08M	15	P-725.08L	P-725.08S
M25 x 0.75	P-725.11M	22	P-725.11L	P-725.11S
W0.8 x 1/36"	P-725.12M	16	P-725.12L	P-725.12S
SM1 (1.035"-40)	P-725.13M	22	P-725.13L	P-725.13S
M34 x 1	P-725.14M	28	P-725.14L	P-725.14S

Refer to "Optional Accessories" (p. 11) for details.

Product View

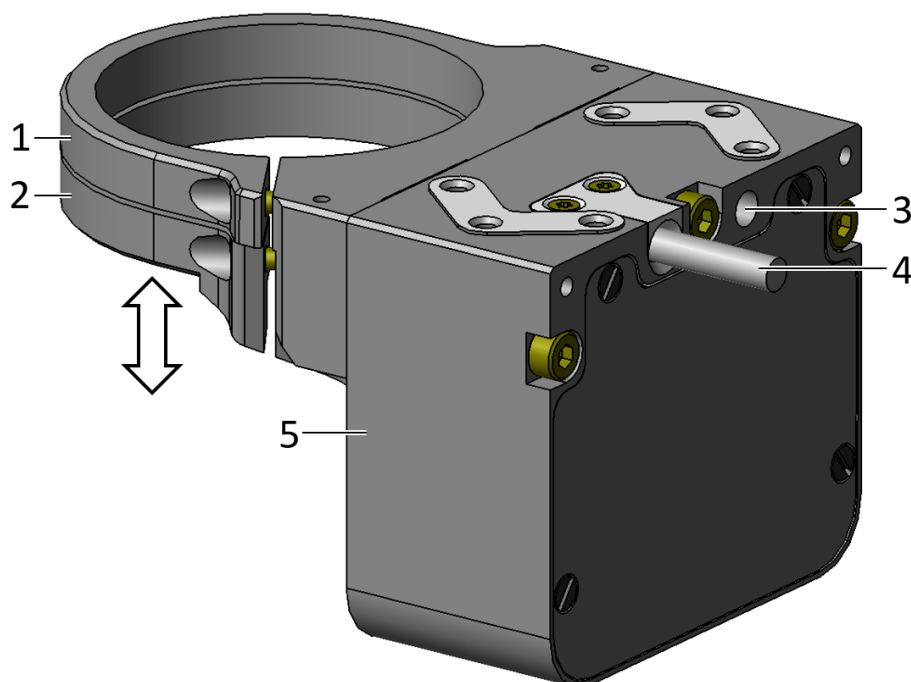


Fig. 1: Example view of a P-725.xCDE1

- 1 Aperture for thread adapter (microscope side clamping ring)
- 2 Aperture for thread adapter (objective side clamping ring)
- 3 M4 hole for connecting the protective earth conductor
- 4 Cable exit
- 5 Base body

Double arrow: Direction of motion of objective side clamping ring

Product Labeling

Labeling	Description
P-725.4CDE1	Product name (example), the characters following the period refer to the model
123456789	Serial number (example), individual for each P-725 Meaning of each position (from the left): 1 = internal information 2 and 3 = manufacturing year 4 to 9 = consecutive numbers
PIFOC	Brand name
Country of origin: Germany	Country of origin
	Warning sign "Pay attention to the manual!"
	Old equipment disposal (p. 28)

Labeling	Description
WWW.PI.WS	Manufacturer's address (website)
	CE conformity mark
	Manufacturer's logo
	Symbol for the protective earth conductor (p. 12), indicates the protective earth connector on the P-725

Scope of Delivery

P-725.xCDE1S system according to order (p. 8), consisting of:

Product no.	Description
P-725.xCDE1	PIFOC focus scanner according to order (p. 8)
E-709.1C1L	Digital piezo controller, 1 axis, -30 to 130 V, capacitive sensor, monitoring functionality, benchtop device
C-501.24050M8	Desktop power adapter, primary 100-240 V AC, secondary 24 V DC 50 W M8
3763	Power cord
000036360	USB cable (type A to mini B) for connecting to the PC, 3 m
000084853	4 adhesive feet for E-709
000036450	M4 screw set for protective earth, consisting of: - One flat-head screw with cross recess, M4x8, ISO 7045 2 lock washers 2 flat washers
000044198	T6 right-angle Torx key for aperture
C-990.CD1	Data storage device with PC software from PI
PZ289EK	Short instructions for E-709.xxG / E-727 / E-754 digital piezo controllers
PZ240EK	Short instructions for P-5xx / P-6xx / P-7xx piezo positioners

Optional Accessories

- To order, contact the customer service department (p. 23).

P-725.xxM Thread Adapters

Product no.	Description
P-725.02M	Thread adapter for microscope, M26 x 0.75, aperture diameter 22 mm
P-725.03M	Thread adapter for microscope, M27 x 0.75, aperture diameter 22 mm
P-725.04M	Thread adapter for microscope, M28 x 0.75, aperture diameter 22 mm
P-725.05M	Thread adapter for microscope, M32 x 0.75, aperture diameter 25 mm
P-725.06M	Thread adapter for microscope, M26 x 1/36", aperture diameter 22 mm
P-725.08M	Thread adapter for microscope, M19 x 0.75, aperture diameter 15 mm
P-725.11M	Thread adapter for microscope, M25 x 0.75, aperture diameter 22 mm
P-725.12M	Thread adapter for microscope, W0.8 x 1/36", aperture diameter 16 mm
P-725.13M	Thread adapter for microscope, SM1 (1.035"-40), aperture diameter 22 mm
P-725.14M	Thread adapter for microscope, M34 x 1, aperture diameter 28 mm

P-725.xxl Thread Adapters

Product no.	Description
P-725.02L	Thread adapter for objective, M26 x 0.75, aperture diameter 22 mm
P-725.03L	Thread adapter for objective, M27 x 0.75, aperture diameter 22 mm
P-725.04L	Thread adapter for objective, M28 x 0.75, aperture diameter 22 mm
P-725.05L	Thread adapter for objective, M32 x 0.75, aperture diameter 25 mm
P-725.06L	Thread adapter for objective, M26 x 1/36", aperture diameter 22 mm
P-725.08L	Thread adapter for objective, M19 x 0.75, aperture diameter 15 mm
P-725.11L	Thread adapter for objective, M25 x 0.75, aperture diameter 22 mm
P-725.12L	Thread adapter for objective, W0.8 x 1/36", aperture diameter 16 mm
P-725.13L	Thread adapter for objective, SM1 (1.035"-40), aperture diameter 22 mm
P-725.14L	Thread adapter for objective, M34 x 1, aperture diameter 28 mm

P-725.xxS Spacers

Product no.	Description
P-725.02S	Objective spacer for correcting the parfocal length, 11.5 mm, M26 x 0.75 mm thread
P-725.03S	Objective spacer for correcting the parfocal length, 11.5 mm, M27 x 0.75 mm thread
P-725.04S	Objective spacer for correcting the parfocal length, 11.5 mm, M28 x 0.75 mm thread

Product no.	Description
P-725.05S	Objective spacer for correcting the parfocal length, 11.5 mm, M32 × 0.75 mm thread
P-725.06S	Objective spacer for correcting the parfocal length, 11.5 mm, M26 × 1/36" thread
P-725.08S	Objective spacer for correcting the parfocal length, 11.5 mm, M19 × 0.75 mm thread
P-725.11S	Objective spacer for correcting the parfocal length, 11.5 mm, M25 × 0.75 mm thread
P-725.12S	Objective spacer for correcting the parfocal length, 11.5 mm, W0.8 × 1/36" thread
P-725.13S	Objective spacer for correcting the parfocal length, 11.5 mm, 1.035"-40 (SM1) thread
P-725.14S	Objective spacer for correcting the parfocal length, 11.5 mm, M34 × 1 mm thread

Unpacking

1. Unpack the P-725 with care.
2. Compare the contents with the scope of delivery according to the contract and the delivery note.
3. If any parts are damaged or missing, contact our customer service department immediately (p. 23).
4. Keep all packaging materials in case the product needs to be returned.

Installing

Connecting the P-725 to the Protective Earth Conductor

INFORMATION

- Pay attention to the applicable standards for connecting the protective earth conductor.

INFORMATION

- If there is any vibration in your application, securely fasten the screw connection for the protective earth conductor in a suitable manner (e.g., with liquid adhesive) to prevent it from loosening. If this is not possible, check the screw connection regularly and tighten the screw if necessary.

There is an M4 hole in the P-725 for connecting the protective earth conductor. This hole is marked with the symbol for the protective earth conductor . Refer to "Dimensions" (p. 26).

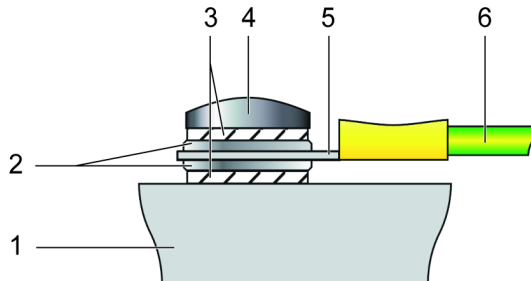


Fig. 2: Protective earth conductor mounting (profile view)

- 1 P-725 base body
- 2 Flat washer
- 3 Lock washer
- 4 Screw
- 5 Cable lug
- 6 Protective earth conductor

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ The P-725 is **not** connected to the electronics.

Tools and accessories

- Suitable protective earth conductor:
 - Cable cross section $\geq 0.75 \text{ mm}^2$
 - Contact resistance $< 0.1 \Omega$ at 25 A at all connection points relevant for mounting the protective earth conductor
- The supplied M4 protective earth conductor screw set for connecting the protective earth conductor (p. 10)
- Suitable screwdriver

Connecting the P-725 to the protective earth conductor

1. If necessary, firmly attach a suitable cable lug to the protective earth conductor.
2. Use the M4 screw (together with the flat and lock washers) to attach the cable lug of the protective earth conductor to the threaded hole in the P-725 as shown in the profile view.
3. Tighten the M4 screw with a torque of 1.2 Nm to 1.5 Nm.

Mounting the P-725

The PIFOC focus scanners are inserted between the microscope turret and the objective using thread adapters (see Fig. 3). In addition, it is possible to mount the P-725 using the mounting holes in its base body (p. 15).

Mounting the P-725 with P-725.xxM/.xxL Thread Adapters to the Microscope

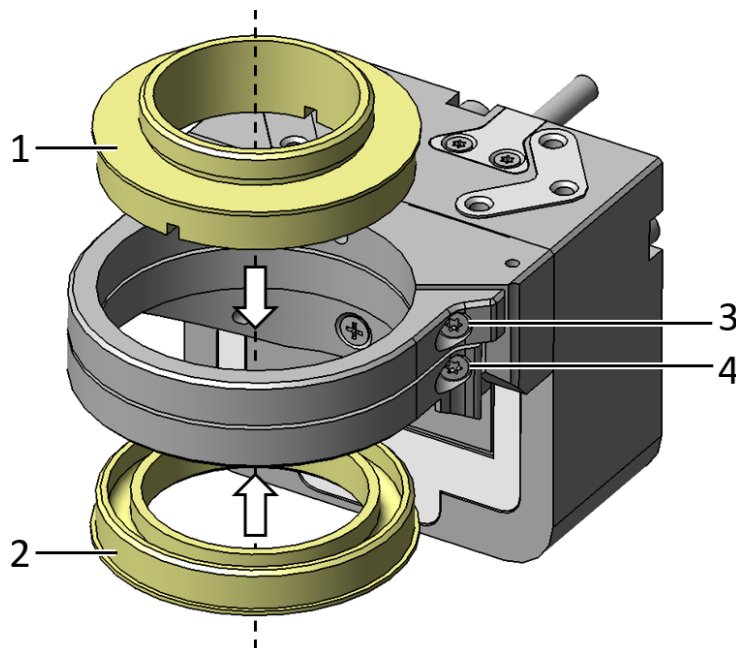


Fig. 3: Schematic diagram of parts relevant for mounting the P-725

- 1 Thread adapter, microscope side
- 2 Thread adapter, objective side
- 3 Screw for thread adapter on the microscope side
- 4 Screw for thread adapter on the objective side

Arrows: Direction for fixing the respective thread adapter into the P-725's aperture

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ The P-725 is **not** connected to the electronics.

Tools and accessories

- P-725.xxM thread adapter, microscope side (p. 11)
(not included in the scope of delivery; refer to "Optional Accessories" (S. 11))
- P-725.xxL thread adapter, objective side (p. 11)
(not included in the scope of delivery; refer to "Optional Accessories" (S. 11))
- Supplied right-angle Torx key (p. 10)
- Key tool (in the scope of delivery of the P-725.xxM thread adapter)

Mounting the P-725 with P-725.xxM/.xxL thread adapters to the microscope

1. Insert the thread adapter for the objective side into the aperture of the P-725 from below.
2. Use the Torx key to tighten the screw for the thread adapter on the objective side (refer to Fig. 3, p. 14).
3. Screw the microscope side thread adapter into the microscope.
4. Tighten the thread adapter on the microscope side with the wide side of the key tool.
5. Insert the P-725 into the thread adapter on the microscope side.

6. Use the Torx key to tighten the screw for the thread adapter on the microscope side (refer to Fig. 3, p. 14).
7. Screw the objective from below into the thread adapter on the objective side of the P-725. Do not exceed the maximum torque of 0.6 Nm.

Mounting the P-725 Using the Mounting Holes in the Base Body (Optional)

NOTICE



Warping of the P-725 when mounted on uneven surfaces!

Mounting the P-725 on an uneven surface could warp the P-725. Warping reduces the accuracy.

- Mount the P-725 on an even surface. The recommended flatness of the surface is $\leq 10 \mu\text{m}$.
- For applications with large temperature fluctuations:
Only mount the P-725 onto surfaces which have the same or similar thermal expansion properties as the P-725 (e.g., an aluminum surface).

INFORMATION

- Note that only the P-725.xxL thread adapter on the objective side is required when mounting the P-725 using the mounting holes in the base body.

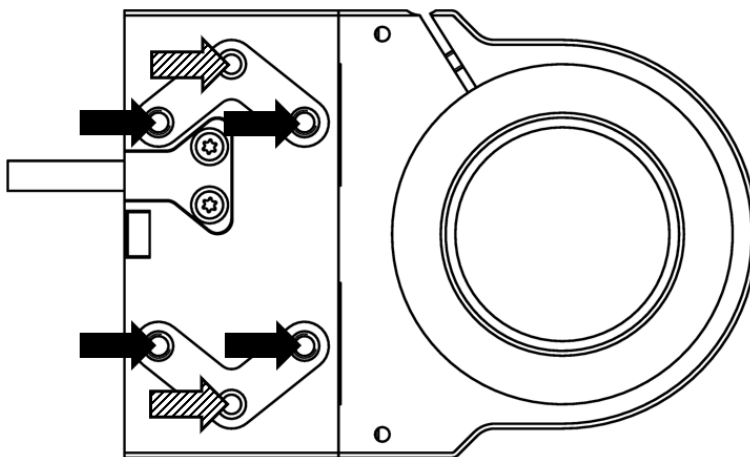


Fig. 4: P-725.xCDE1 (exemplary view from above)

Black arrows: M2.5 mounting holes

Hatched arrows: Locating holes

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ The P-725 is **not** connected to the electronics.
- ✓ You have provided a suitable surface that fulfills the following requirements:
 - The surface has four through-holes for M2.5 screws. Refer to "Dimensions" for the required position and depth of the holes (p. 26).
 - The surface's flatness is $\leq 10 \mu\text{m}$.
 - For applications with large temperature fluctuations: The surface should have the same or similar thermal expansion properties as the P-725 (e.g., an aluminum surface).

Tools and accessories

- Four M2.5 screws of a suitable length (p. 26) for fixing the P-725
- Optional: Two suitable locating pins (p. 26) for aligning the P-725
- P-725.xxL thread adapter, objective side
(not included in the scope of delivery; refer to "Optional Accessories" (S. 11))
- Suitable screwdriver

Mounting the P-725 using the mounting holes in the base body

1. Mount the P-725.xxL thread adapter for the objective side to the P-725:
Clamp the thread adapter in the aperture of the P-725. For details, refer to "Mounting the P-725 with P-725.xxM/.xxL Thread Adapters to the Microscope" (p. 14).
2. Align the P-725 on the surface so that the mounting holes in the base body of the P-725 cover the mounting holes in the surface.
Optional: Use locating pins and the locating holes in the base body of the P-725 for alignment.
3. Insert the screws through the surface into the base body of the P-725.
4. Tighten the screws crosswise. When tightening the screws, pay attention to the specified torque range (p. 27).
5. Screw the objective from below into the thread adapter on the objective side of the P-725. Do not exceed the maximum torque of 0.6 Nm.

Mounting P-725.xxS Spacers

The P-725.xxS spacers are inserted between the microscope turret and the objective to correct the parfocal length.



Fig. 5: P-725.03S spacer (example), inserted between the microscope turret and the objective

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ The P-725 is **not** connected to the electronics.

Tools and accessories

- P-725.xxS spacer
(not included in the scope of delivery, refer to "Optional Accessories" (S. 11))

Mounting P-725.xxS spacers

1. Screw the spacer into the microscope turret.
2. Screw the objective into the spacer.

Connecting the P-725 to the Electronics

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ You have read and understood the user manual for the electronics.
- ✓ The electronics are switched off, i.e., **not** connected to the power supply.

Connecting the P-725 to the electronics

- Connect the plug of the P-725 to the corresponding socket on the electronics (refer to the user manual for the electronics).

Starting and Operating

General Notes on Starting and Operating

CAUTION



Risk of electric shock if the protective earth conductor is not connected!

If the protective earth conductor is not connected or is not connected properly, hazardous touch voltages can occur on the P-725 during system malfunction or failure. Touching the P-725 under these circumstances may result in minor injury due to electric shock.

- Connect the P-725 to a protective earth conductor (p. 12) before use.
- Do **not** disconnect the protective earth conductor during operation.
- If the protective earth conductor needs to be disconnected temporarily (e.g., during modifications), make sure that the P-725 is reconnected to the protective earth conductor before restarting.

NOTICE



Destruction of the piezo actuator due to electric flashovers!

Using the P-725 in environments that increase the electrical conductivity could lead to the destruction of the piezo actuator by electric flashovers. Electric flashovers can be caused by moisture, high humidity, liquids, and conductive materials (e.g., metal dust). In addition, electric flashovers are also possible because of increased conductivity in certain air pressure ranges.

- Avoid operating the P-725 in environments that can increase the electric conductivity.
- Operate the P-725 only under permissible ambient conditions and classifications (p. 25).

NOTICE



Destruction of the piezo actuator due to continuously high voltage!

The constant application of high voltage to piezo actuators can lead to leakage currents and flashovers that will destroy the ceramic.

- If the P-725 is not being used, but the controller is to remain switched on to ensure temperature stability, set the piezo voltage to 0 V on the controller.

NOTICE



Uncontrolled oscillation!

Oscillation can cause irreparable damage to the P-725. Oscillation is indicated by a humming noise and can be caused by the following:

- The load and/or dynamics during operation differ considerably from the calibration settings.
- The P-725 is operated near to its resonant frequency.
- If you notice oscillations, stop the P-725 immediately.

INFORMATION

Systems are calibrated at the factory to achieve optimal positioning accuracy. Replacing the system components could lead to a reduction of position accuracy when positioners are used that have ID chips without calibration data or when LEMO plug connectors are used.

- When connecting the positioner, pay attention to the assignment of the motion axes to the controller channels that is specified by the calibration label on the controller.

If position accuracy is reduced after replacing the P-725 or the controller:

- Recalibrate the axis displacement (refer to the controller manual) or contact our customer service department (p. 23).

Starting and Operating the P-725

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ You have read and understood the general notes on starting and operating (p. 18).
- ✓ You have installed the P-725 correctly (p. 12) and connected it to the electronics (p. 17).
- ✓ You have read and understood the user manual for the electronics (p. 10).

Starting and operating the P-725

- Follow the instructions on starting and operating the P-725 in the user manual for the electronics.

Discharging the P-725

The P-725 must be discharged in the following cases:

- When the P-725 is not used but the electronics remain switched on to ensure temperature stability
- Before demounting (e.g., before cleaning and transporting the P-725) and for modifications
- If the connecting cable of the P-725 is accidentally pulled out of the electronics during operation

Requirements

- ✓ You have read and understood the safety precautions (p. 6).

Discharging a P-725 connected to the electronics

In closed-loop operation:

1. Switch off the servo mode on the electronics.
2. Set the piezo voltage to 0 V on the electronics.

In open-loop operation:

- Set the piezo voltage to 0 V on the electronics.

Discharging a P-725 not connected to the electronics

- Connect the P-725's voltage plug to the **switched-off** electronics from PI for 10 seconds.

Maintenance

NOTICE



Misalignment from loosening screws!

The P-725 is maintenance-free and precision adjusted.

- Loosen screws only when instructed in this user manual.
- Do **not** open the P-725.

Cleaning the P-725

NOTICE



Short-circuiting due to cleaning fluid getting into the housing!

Liquid getting into the housing of the P-725 could lead to the destruction of the piezo actuator by electric flashovers.

- Discharge the P-725 before cleaning (p. 20).
- Disconnect the P-725 from the electronics before cleaning.
- Prevent any cleaning fluid from getting into the housing of the P-725.

NOTICE



Damage from ultrasonic cleaning!

Ultrasonic cleaning can damage the P-725.

- Do **not** do any ultrasonic cleaning.

NOTICE



Damage due to use of unsuitable cleaning agents!

Some cleaning agents may cause damage to the P-725.

- Use mild cleaning agents only (e.g., isopropyl alcohol).

Requirements

- ✓ You have discharged the P-725 (p. 20).
- ✓ You have disconnected the P-725 from the electronics.

Other materials required

- Soft, lint-free cloth
- Mild cleaning agent or disinfectant (e.g., isopropyl alcohol)

Cleaning the P-725

1. Dampen the cloth with the cleaning agent or disinfectant.
2. Wipe the surfaces of the P-725 carefully.

Disassembly of P-725 and Thread Adapter

NOTICE



Parts falling down when removing the thread adapters!

While removing the upper thread adapter (microscope side), the P-725 can drop if it is not secured properly or screwed tightly. While removing the lower thread adapter (objective side), the thread adapter and the screwed in objective might fall.

- Secure all parts from falling when the thread adapters are removed:
 - Screw the objective out before disassembly.
 - While loosening the upper thread adapter from the P-725, hold the P-725 tightly.

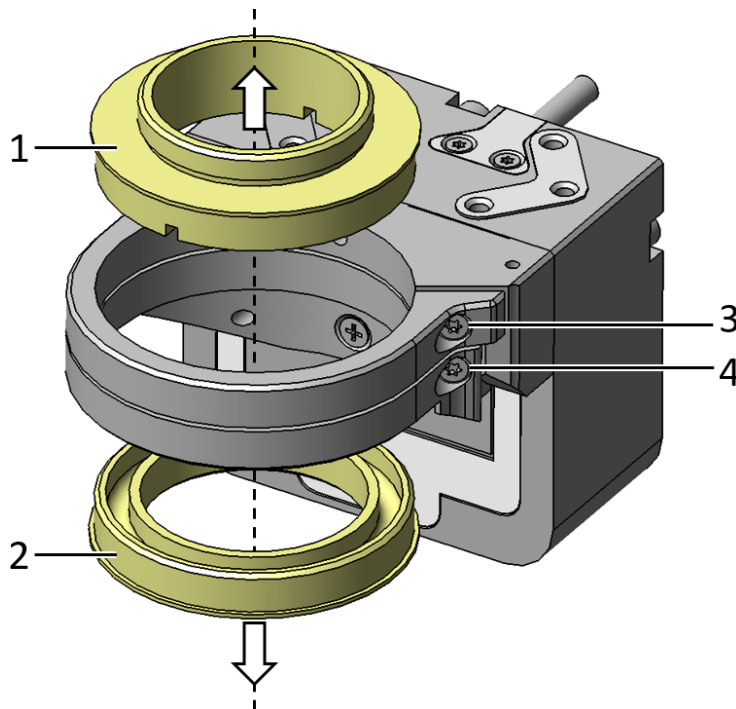


Fig. 6: Schematic diagram of parts relevant for disassembling

- 1 Thread adapter, microscope side
- 2 Thread adapter, objective side
- 3 Screw for thread adapter on the microscope side
- 4 Screw for thread adapter on the objective side

Arrows: Direction for removing the respective thread adapter

Requirements

- ✓ You have read and understood the safety precautions (p. 6).
- ✓ The P-725 is **not** connected to the electronics.

Tools and accessories

- Supplied right-angle Torx key (p. 10)
- Key tool (in the scope of delivery of the P-725.xxM thread adapter)

Disassembly of P-725 and thread adapter

1. Unscrew the objective from the objective side thread adapter of the P-725.
2. Remove the objective side thread adapter: Hold tightly to the objective side thread adapter and use the Torx key to loosen the screw for the objective side thread adapter (refer to Fig. 6, p. 22).
3. Remove the P-725: Hold the P-725 tightly and use the Torx key to loosen the screw for the microscope side thread adapter (refer to Fig. 6, p. 22).
4. Use the wide side of the key tool to unscrew the thread adapter on the microscope side.

Customer Service

For inquiries and orders, contact your PI sales engineer or send us an email (service@pi.de).

- If you have questions regarding your system, provide us with the following information:
 - Product and serial numbers of all products in the system
 - Firmware version of the controller (if applicable)
 - Version of the driver or the software (if applicable)
 - PC operating system (if applicable)
- If possible: Take photographs or make videos of your system that can be sent to our customer service department if requested.

The current versions of the user manuals can be downloaded at www.pi.ws.

Technical Data

Subject to change. You can find the latest product specifications on the product web page at www.pi.ws.

Specifications

Motion	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Active axes	Z	Z		
Travel range in Z	100	400	µm	+20 / -0 %
Travel range in Z, open loop	120	420	µm	±20 %
Linearity error in Z*	0.2	0.2	%	Max.

Positioning	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Minimum incremental motion in Z	2	4	nm	
Point repeatability, 10 % step, 1 Sigma	20	20	nm	Max.
Integrated sensor	Capacitive, direct position measuring	Capacitive, direct position measuring		
Sensor noise, 1 Sigma	5	5	nm	Max.

Drive Properties	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Drive type	Piezo actuator/PICMA®	Piezo actuator/PICMA®		
Electrical capacitance in Z	3.2	6.2	µF	±20 %

Mechanical Properties	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Stiffness in Z	0.23	0.12	N/μm	Min.
Resonant frequency in Z, unloaded	470	230	Hz	Min.
Resonant frequency in Z, under load with 150 g	185	120	Hz	Min.
Permissible push force in Z	100	100	N	Max.
Permissible pull force in Z	20	20	N	Max.
Guide	Flexure guide with lever amplification	Flexure guide with lever amplification		
Overall mass	280	280	g	±5 %
Material	Stainless steel, aluminum	Stainless steel, aluminum		
Objective diameter	39	39	mm	Max.

Miscellaneous	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Operating temperature range	P-725.xCDE1: 10 to 70 E-709.1C1L: 5 to 40	P-725.xCDE1: 10 to 70 E-709.1C1L: 5 to 40	°C	
Connector	D-sub 7W2 (m)	D-sub 7W2 (m)		
Cable length	1.5	1.5	m	+50/-0 mm

Controller	P-725.1CDE1S	P-725.4CDE1S	Unit	Tolerance
Controller type	E-709.1C1L	E-709.1C1L		
Communication interfaces	RS-232, SPI, USB	RS-232, SPI, USB		
I/O Lines	4 digital inputs, 4 digital outputs, 1 servo cycle output, 1 reset input (each TTL). I/O for external synchronization of the servo cycle (100 kHz). RxD and TxD for RS-485.	4 digital inputs, 4 digital outputs, 1 servo cycle output, 1 reset input (each TTL). I/O for external synchronization of the servo cycle (100 kHz). RxD and TxD for RS-485.		
Command set	GCS 2.0	GCS 2.0		
User software	PIMikroMove	PIMikroMove		
Software drivers	GCS DLL (with code examples for the most common programming languages such as C++, C#, VB.NET, Python, Delphi), driver for LabVIEW, MATLAB library	GCS DLL (with code examples for the most common programming languages such as C++, C#, VB.NET, Python, Delphi), driver for LabVIEW, MATLAB library		
Application-related functions	Data recorder	Data recorder		
Controller dimensions	160 mm × 104 mm × 44 mm	160 mm × 104 mm × 44 mm		

* The specified value can only be reached with recommended digital controllers.

Position noise: 1 sigma

The objective is not included in the scope of delivery.

Maximum Ratings

P-725 focus scanners are designed for the following operating data:

Model	Maximum operating voltage 	Maximum operating frequency (unloaded) ¹ 	Maximum power consumption ² 
P-725.1CDE1	-20 to 120 V	157 Hz	8.6 W
P-725.4CDE1	-20 to 120 V	77 Hz	17.4 W

¹ To ensure stable operation, the maximum operating frequency has been defined as around one third of the mechanical resonant frequency.

² The heat that is generated by the piezo actuator during dynamic operation limits the value for maximum power consumption.

Details can be found at the following website:

<https://www.physikinstrumente.com/en/technology/piezo-technology/properties-piezo-actuators/electrical-operation/>

Ambient Conditions and Classifications

Pay attention to the following ambient conditions and classifications for the P-725:

Area of application	For indoor use only
Maximum altitude	2000 m above msl
Air pressure	1070 to 780 hPa
Relative humidity	Highest relative humidity 80 % for temperatures up to 31 °C Decreasing linearly to 50 % relative humidity at 40 °C
Operating temperature*	10 to 70 °C
Storage temperature	-20 to 60 °C
Transport temperature	-20 to 60 °C
Overvoltage category	II
Protection class	I
Degree of pollution	1
Degree of protection according to IEC 60529	IP20

* The specifications apply to room temperature (22 °C ± 3 °C). Outside of this range, the specifications can deviate.

Dimensions

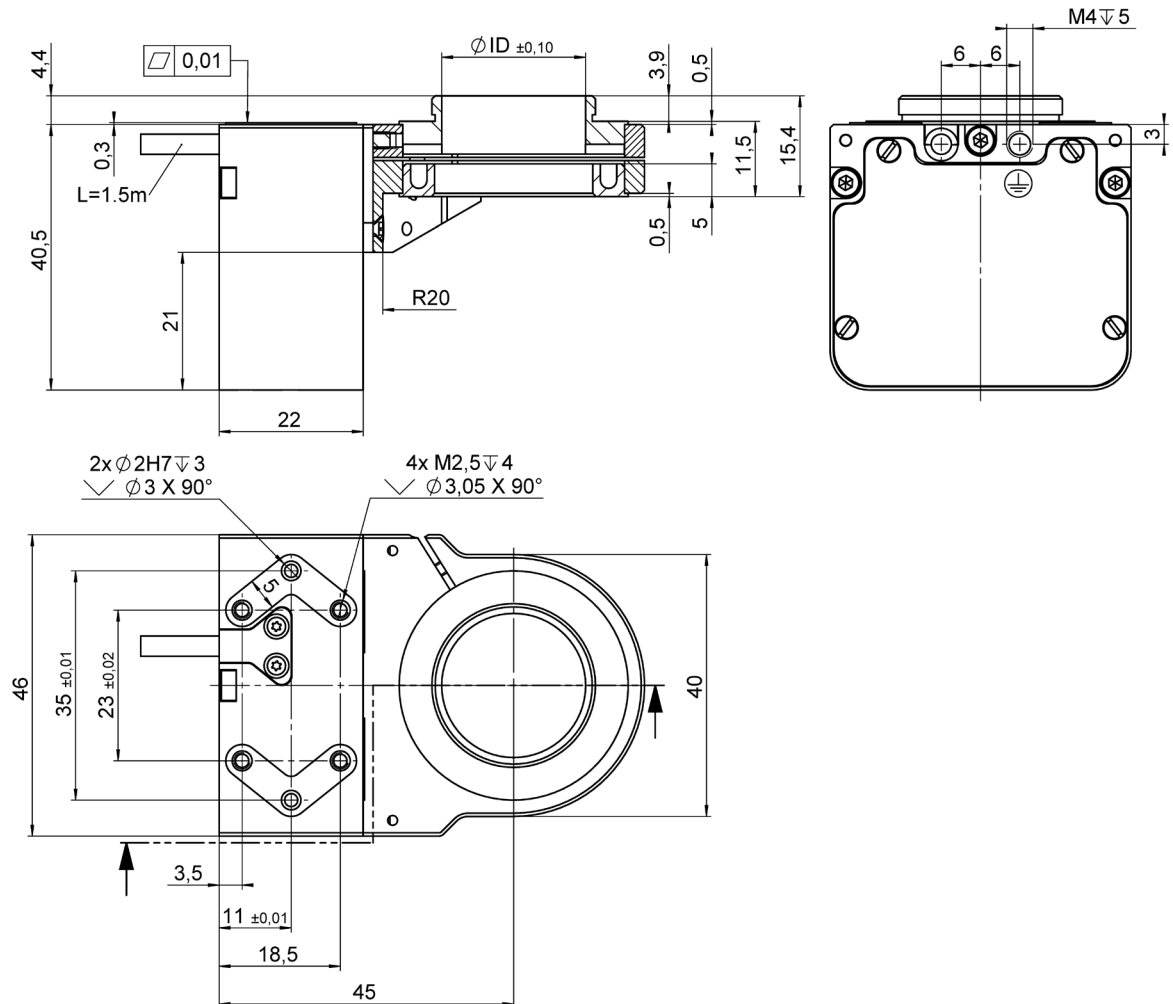


Fig. 7: P-725.1CDE1 and P-725.4CDE1, dimensions in mm (please order adapters and spacers separately)

Torque for Stainless Steel Screws (A2-70)

Screw size	Minimum torque	Maximum torque
M6	4 Nm	6 Nm
M5	2.5 Nm	3.5 Nm
M4	1.5 Nm	2.5 Nm
M3	0.8 Nm	1.1 Nm
M2.5	0.3 Nm	0.4 Nm
M2	0.15 Nm	0.2 Nm
M1.6	0.06 Nm	0.12 Nm

- Pay attention to the screw-in depth required for the respective material in accordance with VDI directive 2230.

Pin Assignment

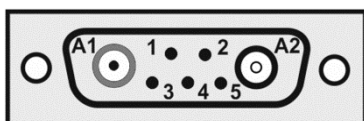


Fig. 8: D-sub 7W2 (m) connector (contact side)

Pin	Signal	Function
A1 inner conductor	Input	Piezo voltage +
A2 inner conductor	Output	Probe sensor signal (immovable part of the capacitive sensor)
A2 outer conductor	GND	Shield
1	Bidirectional	Data line for ID chip
2	GND	Shield of Target Ground of ID chip when switched on
3	Input	Piezo voltage –
4	---	Not connected
5	Input	Target sensor signal (movable part of the capacitive sensor)

The connector shell is connected to the cable shield.

Old Equipment Disposal

In accordance with EU law, electrical and electronic equipment may not be disposed of in EU member states via the municipal residual waste.

Dispose of your old device according to international, national, and local rules and regulations.

To fulfill the responsibility as the product manufacturer, Physik Instrumente (PI) SE & Co. KG undertakes environmentally correct disposal of all old PI equipment made available on the market after 13 August 2005 without charge.

Any old PI device can be sent free of charge to the following address:

Physik Instrumente (PI) SE & Co. KG
Auf der Römerstraße 1
76228 Karlsruhe, Germany

